

Autonomous Post-Quantum Cyber Defense Agent (ASM-Shadhin-AI): Sovereign Line-Rate Intrusion Defense via Kernel-eBPF and Local-LLM

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APPENDIX: SYSTEM INTEGRITY & REPRODUCIBILITY

To establish empirical reproducibility prior to publication, the core subsystems were evaluated using the automated test suite (scripts/test_system_integrity.py) on an authentic Ubuntu Server 22.04 LTS host (Linux kernel 6.8.0, libbpf). Table VIII itemizes the verification criteria and outcomes. All eleven module-level checks passed without manual intervention.

TABLE VIII
SYSTEM DIAGNOSTIC AND LOGICAL INTEGRITY VERIFICATION

#	Target Subsystem	Verification Standard / Criteria	Verdict
1	Shell Automation	POSIX / Bash static syntax validation (bash -n)	PASSED [✓]
2	Local LLM Engine	Modelfile parameters & strict JSON grammar	PASSED [✓]
3	Post-Quantum Crypto	NIST FIPS 203 (ML-KEM-1024) & ML-DSA-65	PASSED [✓]
4	eBPF Kernel Driver	24-byte struct alignment & TTL expiry engine	PASSED [✓]
5	Threat Parser	Suricata alert parser & deterministic XDP fallback	PASSED [✓]
6	AI-Tarpit Deception	Context-exhaustion payloads & honey-tokens	PASSED [✓]
7	Systemd Daemons	Unit syntax for sec-monitor, sec-tarpit, sec-bridge	PASSED [✓]
8	Moving Target Defense	HMAC-SHA256 polymorphic port hopping & grace	PASSED [✓]
9	Encrypted Traffic	Shannon byte entropy ($H \geq 7.1$) & C2 beacon detector	PASSED [✓]
10	Forensic Audit Chain	NIST FIPS 180-4 SHA-512 immutable tamper detection	PASSED [✓]
11	Memory-Hard Auth Guard	RFC 9106 Argon2id (64 MiB) & HMAC-SHA512	PASSED [✓]

*All 11/11 tests passed in production host environment. Full test logs and automated suite are verifiable at:
<https://github.com/Sadek-Mahmud/asm-shadhin-ai>